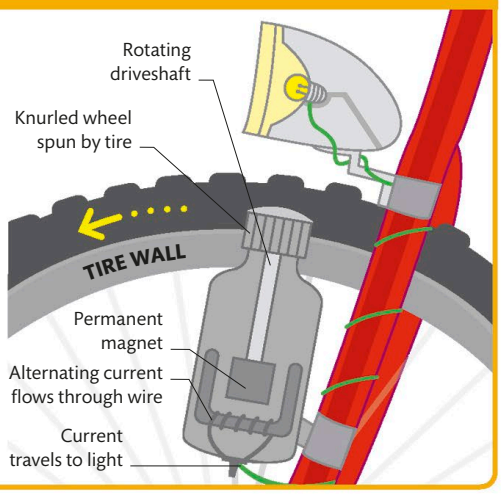
BICYCLE DYNAMOS

A bicycle dynamo powers an electric light through the action of a spinning knurled wheel, which turns because it is connected to the moving tire wall. The spinning wheel rotates a driveshaft attached to a permanent magnet. As this magnet turns, its magnetic field changes, inducing an electric current in the wire coil of the dynamo's electromagnet.

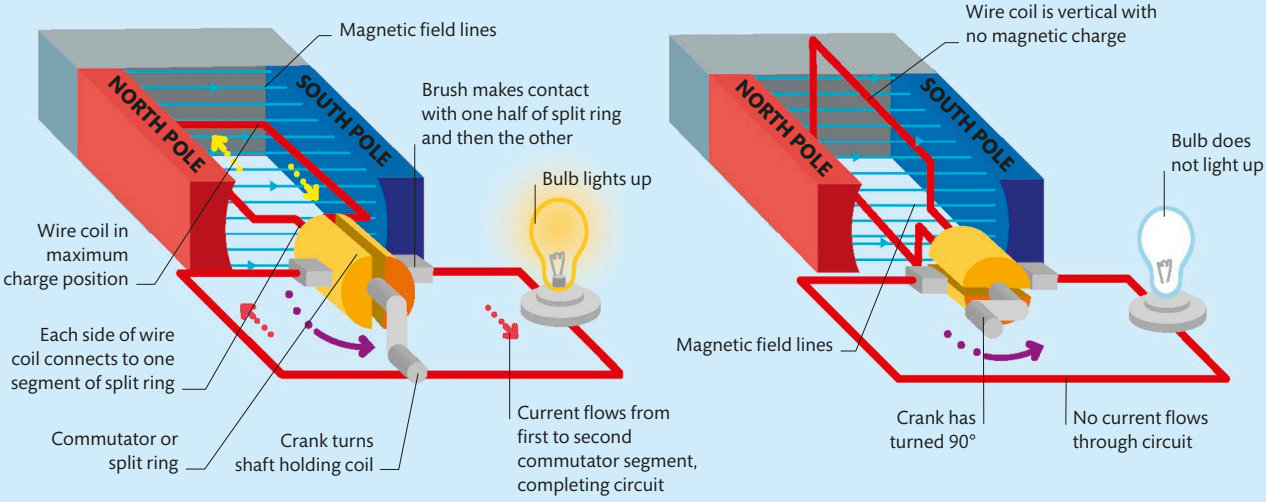


WHAT IS THE FREQUENCY OF AN ALTERNATING CURRENT?

It is how often AC changes direction, measured in hertz (Hz). One Hz is one change per second. Electricity is generated at 60 Hz in the US and often 50 Hz in Europe.

DC generators

DC generators use a device called a commutator to convert AC to DC. It is split into two segments insulated from one another so that no electricity flows between them. The commutator keeps the current flowing in one direction to the output circuit by switching polarity at the same moment that the AC signal reverses direction.



1 Reversing the connection
At peak position, the current flows to the first split ring segment, through the circuit to the second split ring segment, and into the coil, completing the circuit. When the coil turns another 180°, the brush breaks contact with the first segment, making contact with the second segment, on the opposite side of the circuit. The current flows the same way for both the coil's first and second 180° turns.

2 Inconsistent current
When the coil is vertical and not cutting through the horizontal magnetic field lines, no current is produced. This means DC electricity is produced in pulses rather than a steady flow. Most practical DC generators solve this problem by containing multiple coils (so one is always in a horizontal position when the other coils are in less optimal positions) and additional commutators.